

Bank Term Deposit Subscription Prediction

Final Evidence-Based Report - Random Forest vs XGBoost

Purpose: answer each project requirement with a clear conclusion, the reason behind that conclusion, and a readable source reference to the corresponding notebook.

Notebook reference names used throughout this report:

01_Bank_Marketing_EDA.ipynb

02_Random_Forest_Modeling.ipynb

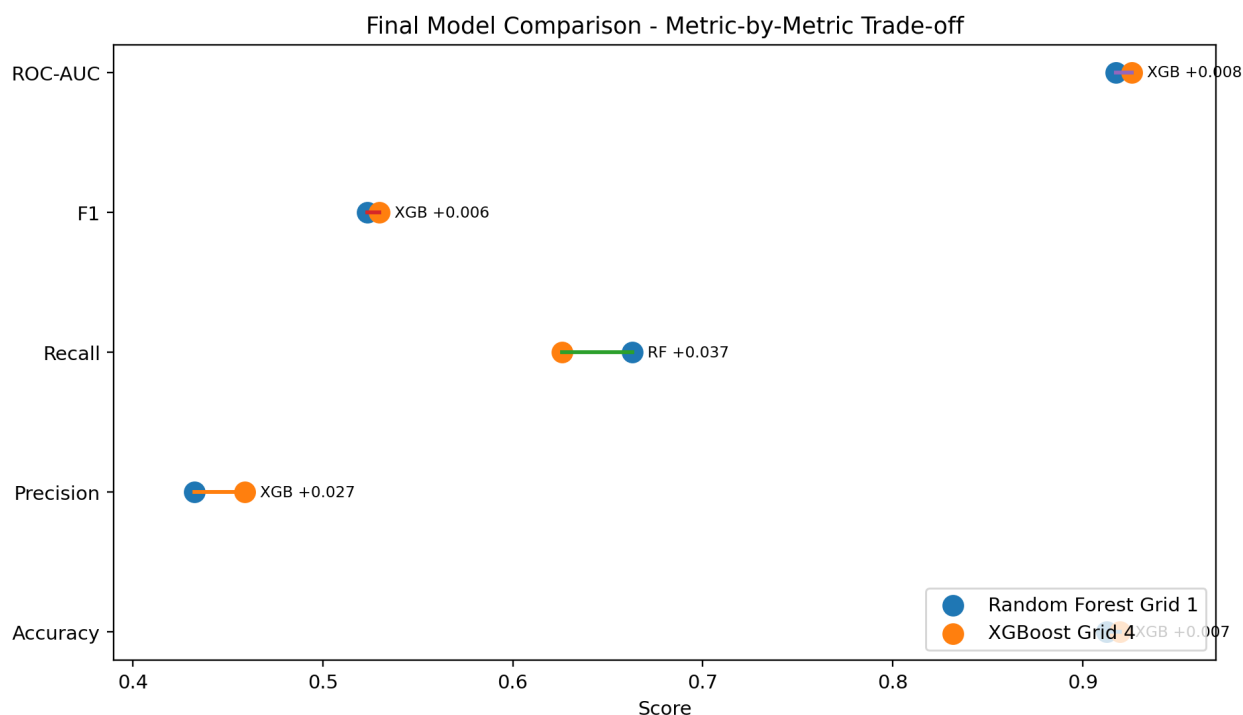
03_XGBoost_Modeling.ipynb

04_Model_Comparison_and_Business_Analysis.ipynb

1. Executive Summary

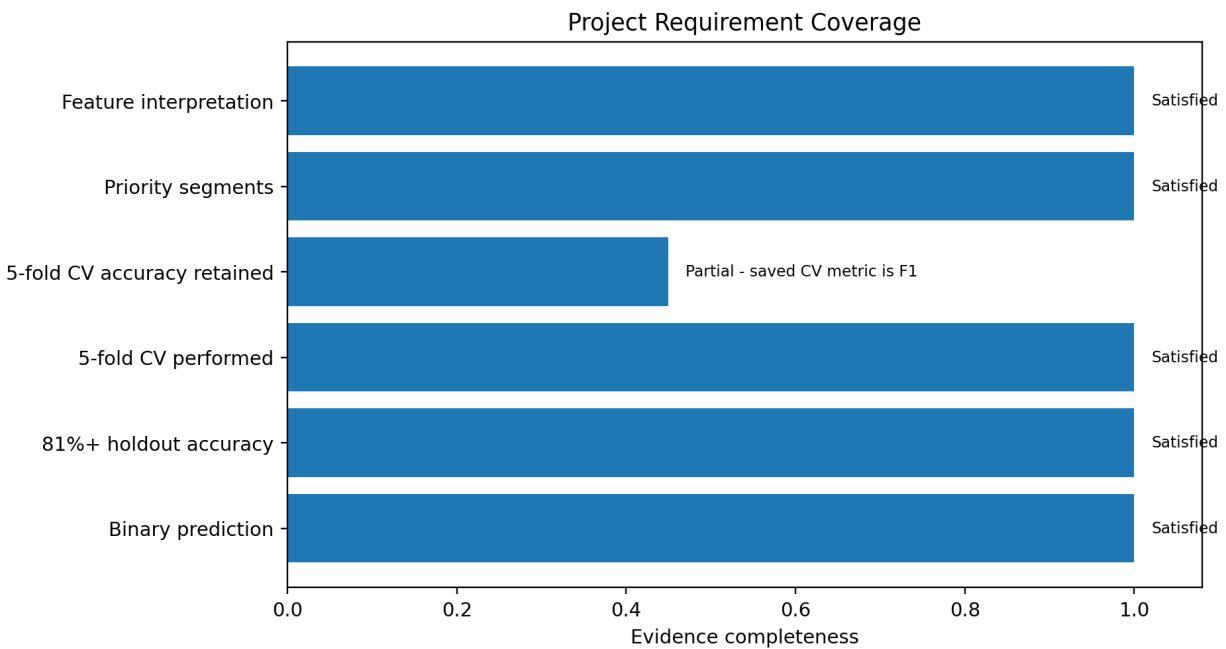
Business Question	Final Answer	Primary Evidence	Notebook Reference
Can subscription be predicted?	Yes	XGBoost Grid 4: 91.95% accuracy; 0.9256 ROC-AUC	03_XGBoost_Modeling.ipynb
Best overall documented model?	XGBoost Grid 4	Wins Accuracy, Precision, F1, ROC-AUC and CV F1; RF wins Recall	04_Model_Comparison_and_Business_Analysis.ipynb
Who should be prioritized?	Multi-factor targeting	Higher observed response in 60+, students/retired, higher balance and Q1 segments	01_Bank_Marketing_EDA.ipynb
What should the bank focus on?	Pre-call profile + financial + timing features	Age, profile, loan/balance and timing are predictive; duration is not available pre-call	03_XGBoost_Modeling.ipynb

Why XGBoost is the main answer: XGBoost Grid 4 gives the strongest overall documented balance. Random Forest detects a slightly larger share of actual subscribers (66.30% recall vs 62.60%), but XGBoost performs better on accuracy, precision, F1 and ROC-AUC. Therefore, XGBoost is the better all-round reference model, while Random Forest remains relevant when recall is the dominant business priority.



Source: 02_Random_Forest_Modeling.ipynb - Balanced GridSearch test performance; 03_XGBoost_Modeling.ipynb - GridSearch 4 test performance; 04_Model_Comparison_and_Business_Analysis.ipynb - final comparison.

2. Project Requirements - Status and Evidence



Requirement	Status	Why this status?	Reference
Predict Yes/No	Satisfied	Both Random Forest and XGBoost were evaluated as binary classifiers.	02_Random_Forest_Modelin g.ipynb / 03_XGBoost_Modeling.ipynb
81%+ accuracy	Satisfied on holdout	XGBoost Grid 4 = 91.95%; RF Grid 1 = 91.26%.	04_Model_Comparison_and_ Business_Analysis.ipynb
5-fold cross-validation	Satisfied	5-fold CV was used during tuning.	02_Random_Forest_Modelin g.ipynb / 03_XGBoost_Modeling.ipynb
Average 5-fold CV accuracy	Not directly retained	Saved CV outputs report F1, not accuracy. It should not be relabeled as CV accuracy.	02_Random_Forest_Modelin g.ipynb / 03_XGBoost_Modeling.ipynb
Priority customer segments	Satisfied	EDA conversion rates identify stronger observed segments.	01_Bank_Marketing_EDA.ipy nb
Feature focus / interpretability	Satisfied	XGBoost feature importance plus EDA provide predictive and directional evidence.	03_XGBoost_Modeling.ipynb / 01_Bank_Marketing_EDA.ip ynb

Important: the only unresolved documentation gap is the exact average 5-fold CV accuracy. The saved outputs support 81%+ holdout accuracy and saved 5-fold CV F1, but not a saved mean CV accuracy value.

3. Why Accuracy Alone Is Not Enough

- The dataset contains 40,000 observations and 14 original attributes.
- The saved EDA reports no missing values and no duplicate rows.
- Only 7.24% of customers subscribed, while 92.76% did not.

Reasoning: because the target is highly imbalanced, a model can achieve high accuracy by favoring the majority 'No' class. This is why the final decision also uses Precision, Recall, F1 and ROC-AUC.

Source: 01_Bank_Marketing_EDA.ipynb - Dataset Information, Missing Values, Duplicate Rows and Target Distribution.

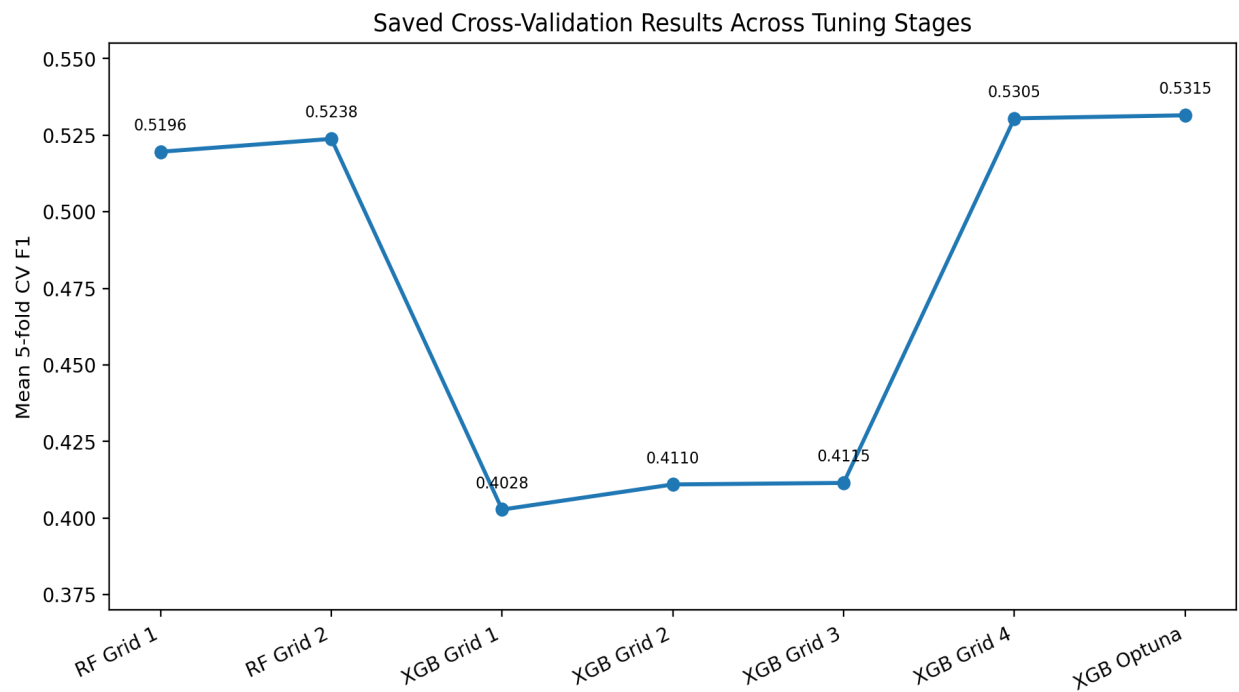
4. Random Forest vs XGBoost - Why XGBoost Was Selected

Metric	Random Forest Grid 1	XGBoost Grid 4	Interpretation
Accuracy	91.26%	91.95%	XGBoost slightly better overall classification
Precision	43.24%	45.91%	XGBoost has fewer incorrect positive predictions proportionally
Recall	66.30%	62.60%	Random Forest captures more actual subscribers
F1 Score	52.34%	52.97%	XGBoost has slightly better precision-recall balance
ROC-AUC	0.9175	0.9256	XGBoost separates the two classes better overall
Mean 5-fold CV F1	0.5196	0.5305	XGBoost has stronger saved CV F1

Conclusion: XGBoost Grid 4 is selected because it wins most of the documented metrics and has the highest F1 and ROC-AUC among the two final candidates. Random Forest is not 'bad'; it is stronger on recall and may be preferred if missed buyers are much more costly than extra calls.

Source: 02_Random_Forest_Modeling.ipynb - Balanced GridSearch 1 evaluation; 03_XGBoost_Modeling.ipynb - GridSearch 4 evaluation; 04_Model_Comparison_and_Business_Analysis.ipynb - Final Model Comparison.

5. Cross-Validation and Tuning Evidence



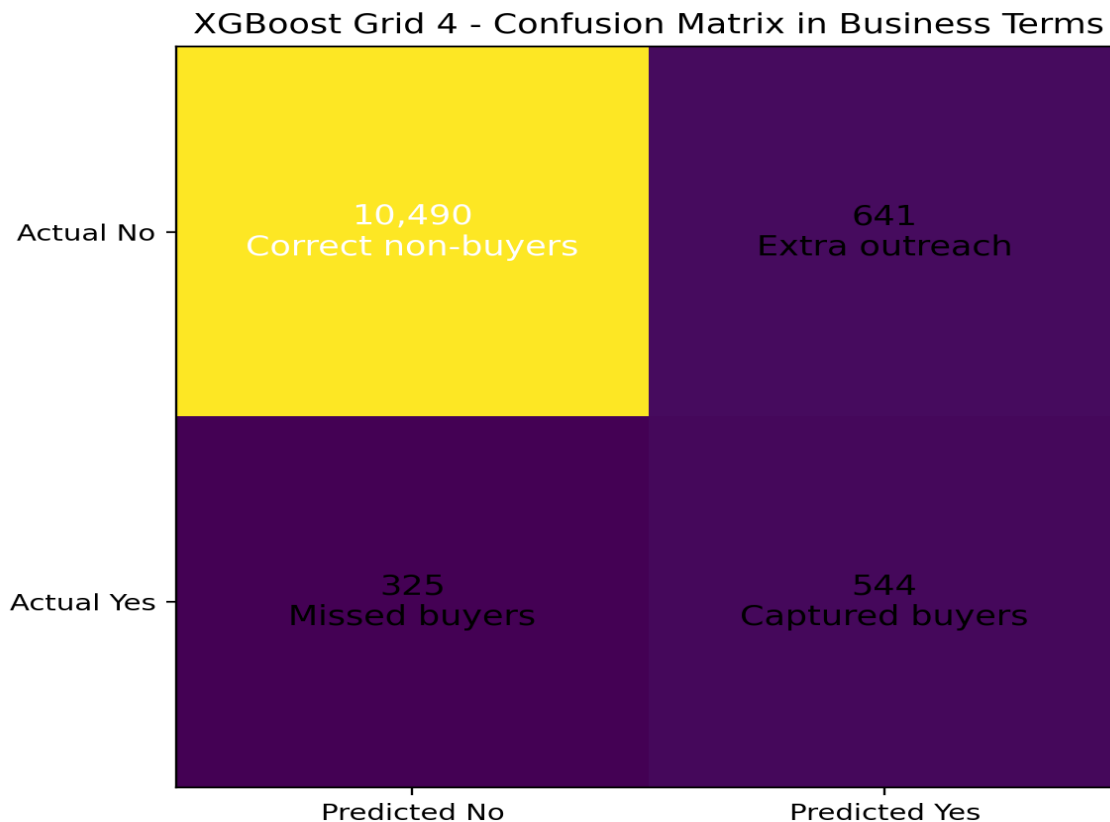
What the tuning shows: XGBoost Grid 4 reached a mean 5-fold CV F1 of 0.5305. Optuna improved this only slightly to 0.5315. This small difference suggests that Grid 4 was already close to the best region found by the wider Optuna search.

Why Grid 4 remains the final report model: its full holdout evaluation is documented and directly comparable in the final analysis, while the saved Optuna evidence primarily establishes the best CV F1.

Success metric caveat: the assignment asks for average 5-fold CV accuracy. The retained outputs report average CV F1 instead. The correct report response is to state this limitation rather than present 0.5305 or 0.5315 as accuracy.

Source: 02_Random_Forest_Modeling.ipynb - GridSearch CV outputs; 03_XGBoost_Modeling.ipynb - GridSearch and Optuna outputs.

6. Final Prediction Performance - Business Meaning

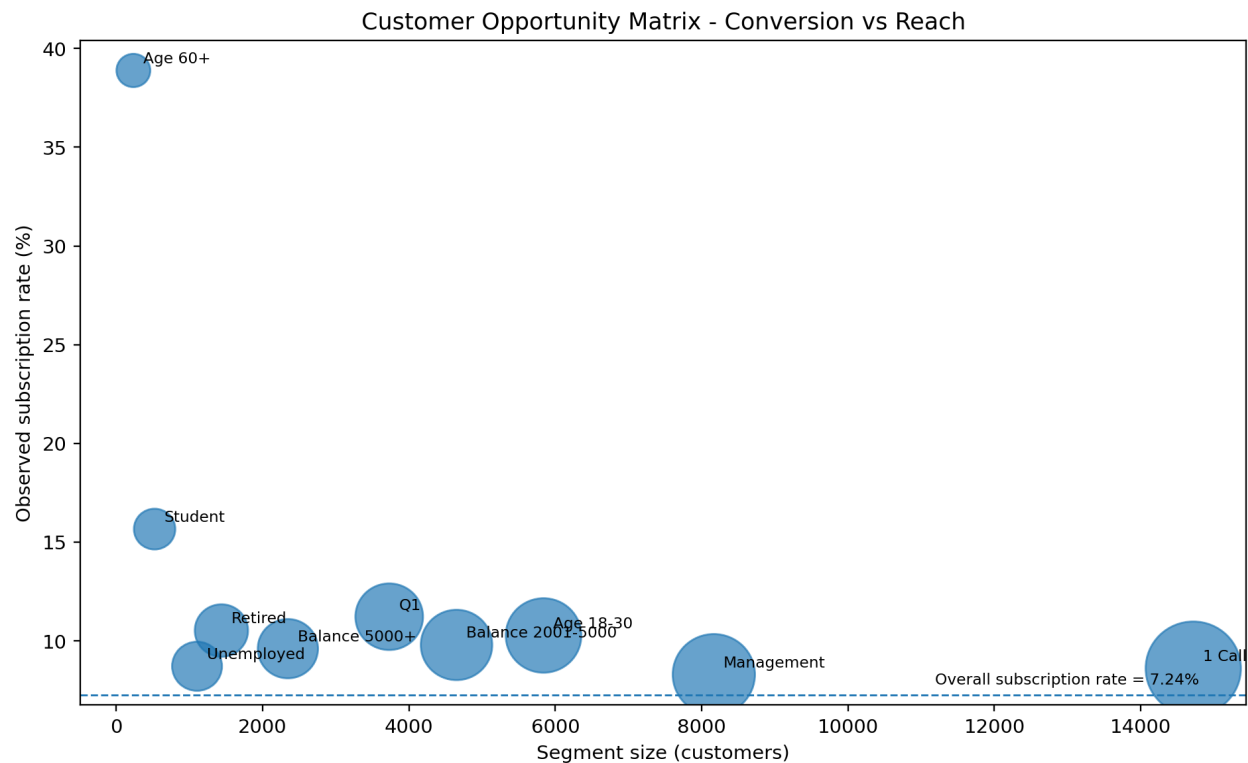


Why this matters: the model correctly captured 544 actual subscribers, but missed 325. It also generated 641 false-positive predictions, which represent extra outreach that would not convert.

Business interpretation: Recall represents how many real buyers the bank captures. Precision represents how efficient positive targeting is. The model should therefore be used to prioritize customers, not to guarantee that a predicted customer will buy.

Source: 04_Model_Comparison_and_Business_Analysis.ipynb - XGBoost Grid 4 Confusion Matrix and Final Prediction Analysis.

7. Which Customer Segments Should Be Prioritized?

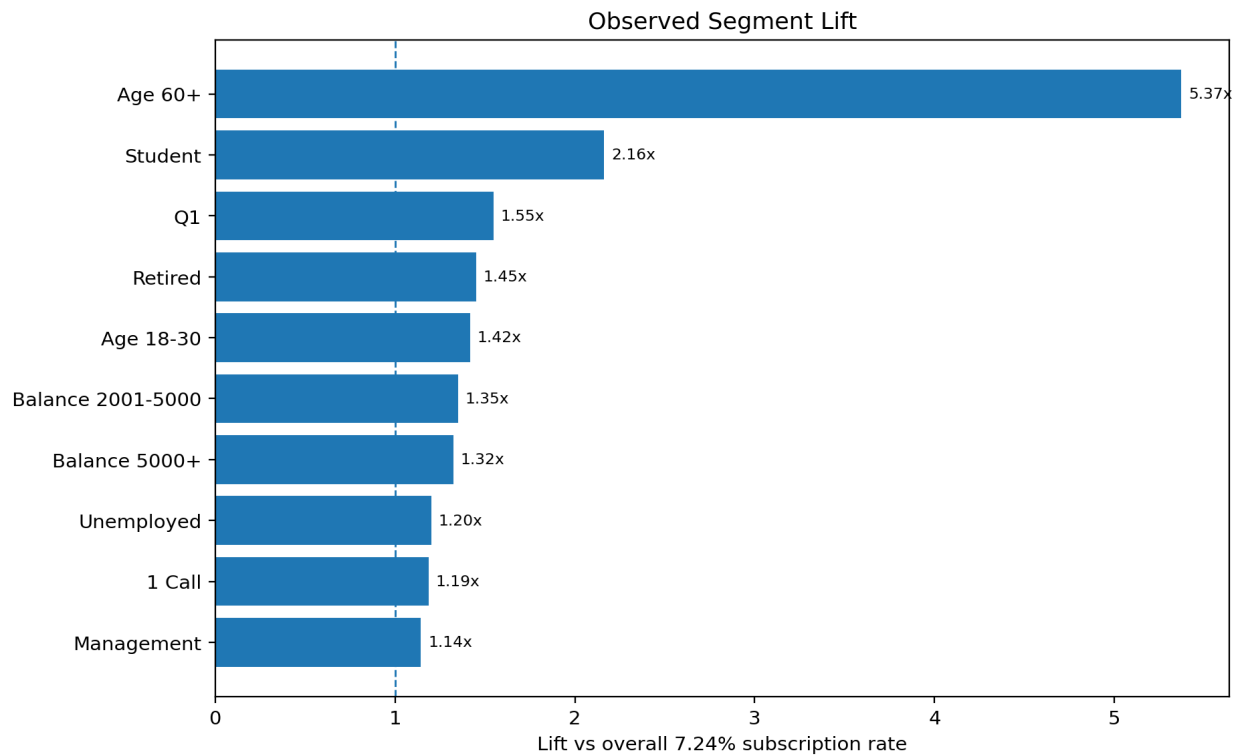


Why this chart is more useful than ranking percentages alone: a segment can have a high conversion rate but be too small to create large campaign volume. Age 60+ has a very high observed rate (38.89%) but only 234 customers, while Management has a lower rate (8.28%) but much greater reach.

Recommended answer: do not target a single demographic rule. Prioritize customers using predicted probability, then use high-response EDA segments to guide campaign design and messaging. Strong observed groups include 60+, students, retired customers, higher-balance customers and Q1 contacts.

Source: 01_Bank_Marketing_EDA.ipynb - Age Group, Job, Balance and Quarter subscription analyses; segment sizes from saved notebook outputs.

8. How Much Better Are the Strong Segments Than Average?

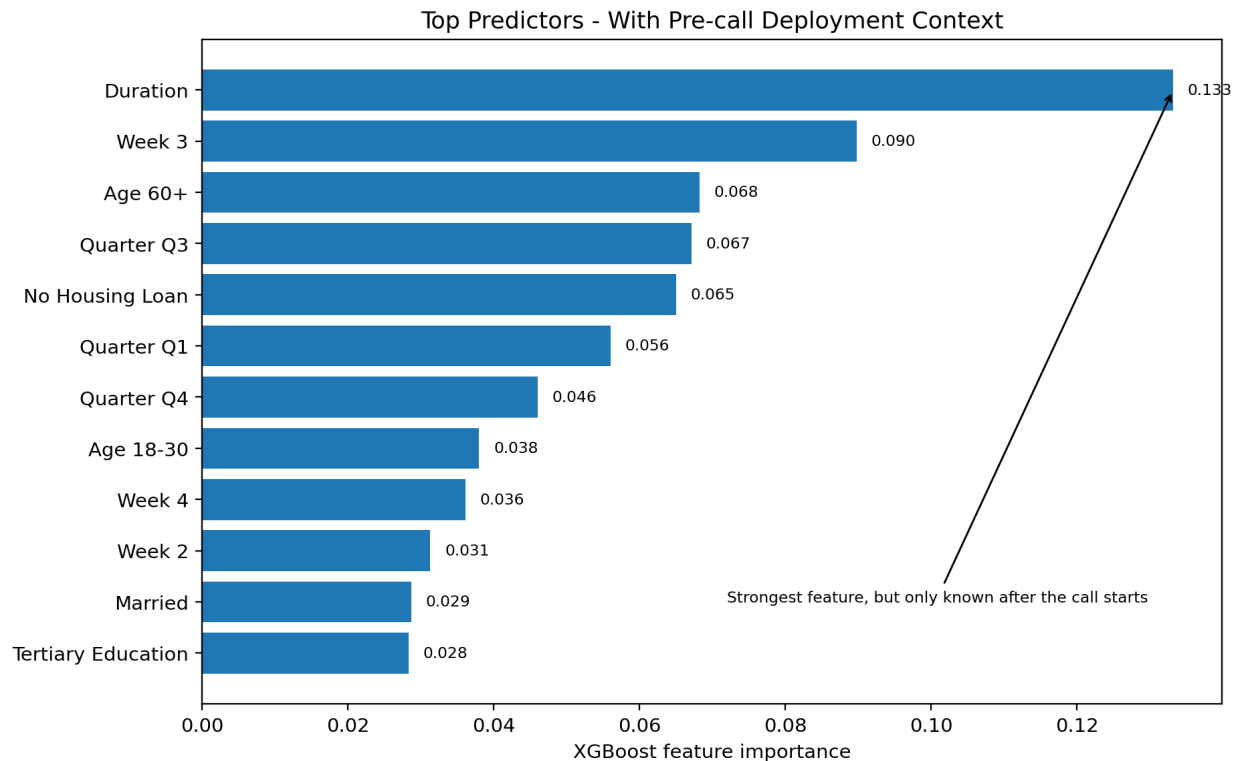


Reasoning: the overall subscription rate is 7.24%. Expressing each segment as a lift over this baseline makes the business difference easier to understand. For example, the observed 60+ rate is more than five times the overall rate, while students are a little over twice the overall rate.

Caution: these are descriptive associations from the EDA. They do not prove that age, job or quarter causes subscription.

Source: 01_Bank_Marketing_EDA.ipynb - Target Distribution plus Age Group, Job and Quarter subscription analyses.

9. What Makes Customers Buy - and Which Features Should We Focus On?



Model evidence: Duration is the strongest individual XGBoost feature (0.1332). Timing, age, housing-loan status, education, marital status, occupation and balance also contribute to prediction.

Why duration cannot be the main targeting answer: duration is only known after the call begins. It is excellent for explaining historical outcomes, but it cannot help decide which customer should receive the call in the first place.

Final feature recommendation: for pre-call targeting, focus on customer age/profile, balance and loan status, together with campaign timing. A production pre-call model should be retrained and validated without duration.

Source: 03_XGBoost_Modeling.ipynb - Top 20 Feature Importance; 01_Bank_Marketing_EDA.ipynb - segment conversion analyses.

10. Final Findings and Business Recommendations

This section directly answers the business questions in the project brief. Each conclusion is supported by the saved model results and EDA evidence, followed by an explanation of why that evidence supports the answer.

10.1 Can the bank predict whether a customer will subscribe?

Answer: Yes. The strongest documented final model, XGBoost Grid 4, achieved 91.95% holdout accuracy, 45.91% precision, 62.60% recall, 52.97% F1 and 0.9256 ROC-AUC.

Why this supports the answer: accuracy shows that most observations were classified correctly, but the dataset is highly imbalanced, with only 7.24% subscribers. For that reason, the answer cannot rely on accuracy alone. Recall of 62.60% shows that the model identifies a meaningful share of actual subscribers, while the F1 score confirms a workable balance between finding subscribers and limiting incorrect positive predictions. ROC-AUC of 0.9256 further shows strong separation between subscribers and non-subscribers.

Business meaning: the prediction should be used as a ranking and prioritization signal for call-center outreach. A positive prediction is not a guaranteed sale; it identifies a customer who deserves relatively higher attention.

Evidence source: 03_XGBoost_Modeling.ipynb - GridSearch 4 Test Performance; 01_Bank_Marketing_EDA.ipynb - Target Distribution.

10.2 Was the 81% success target achieved?

Answer: The 81% threshold is exceeded on the documented holdout evaluation, but the exact requested 5-fold CV accuracy is not retained in the saved outputs. XGBoost Grid 4 reached 91.95% holdout accuracy and Random Forest Grid 1 reached 91.26%.

Why this distinction matters: the project brief specifically asks for 81% or higher accuracy using 5-fold cross-validation and reporting the average score. The notebooks clearly show 5-fold cross-validation during tuning, but the saved average CV metric is F1 rather than accuracy. Therefore, it would be incorrect to present 0.5305 or 0.5315 as CV accuracy.

Conclusion for the requirement: model performance comfortably exceeds 81% on holdout accuracy, and 5-fold CV was performed, but the exact average 5-fold CV accuracy remains a documentation gap that should be recorded or rerun before claiming full compliance with that wording.

Evidence source: 02_Random_Forest_Modeling.ipynb - GridSearch CV and test evaluation; 03_XGBoost_Modeling.ipynb - GridSearch/Optuna CV and test evaluation.

10.3 Which model should be selected?

Answer: XGBoost Grid 4 is the best overall documented reference model. It outperforms Random Forest Grid 1 on accuracy (91.95% vs 91.26%), precision (45.91% vs 43.24%), F1 (52.97% vs 52.34%), ROC-AUC (0.9256 vs 0.9175) and saved CV F1 (0.5305 vs 0.5196). Random Forest is better on recall (66.30% vs 62.60%).

Why XGBoost is preferred: the project requires a robust system rather than optimization of a single metric. XGBoost gives the stronger overall balance and slightly better ability to separate the two classes. Its higher precision also means a somewhat larger proportion of customers flagged as likely subscribers are true subscribers.

When Random Forest could still be preferred: if the bank decides that missing a potential buyer is much more expensive than making extra calls, recall becomes the dominant metric. In that scenario, Random Forest's 66.30% recall can be operationally attractive.

Final model decision: use XGBoost Grid 4 as the project reference model; keep Random Forest as the recall-oriented alternative.

Evidence source: 02_Random_Forest_Modeling.ipynb - Balanced GridSearch 1 Evaluation; 03_XGBoost_Modeling.ipynb - GridSearch 4 Evaluation; 04_Model_Comparison_and_Business_Analysis.ipynb - Final Model Comparison.

10.4 Which customers should the client prioritize?

Answer: Prioritization should be multi-factor and probability-based, not based on one demographic rule. The EDA shows several segments with above-average observed subscription rates: customers aged 60+, students and retired customers, higher-balance customers, customers without a housing loan, and Q1 contacts.

Why one segment is not enough: the 60+ group has a very high observed subscription rate (38.89%) but is relatively small. A larger segment can create more total opportunities even with a lower conversion rate. Therefore, campaign value depends on both *propensity* and *reach*.

Recommended targeting approach: first rank customers using the model's predicted probability. Then use EDA segments to define campaign tiers, messaging and contact timing. High-probability customers who also belong to historically stronger-response groups should receive the highest priority.

Evidence source: 01_Bank_Marketing_EDA.ipynb - Age Group, Job, Balance, Housing and Quarter analyses; 04_Model_Comparison_and_Business_Analysis.ipynb - Customer Opportunity Analysis.

10.5 What makes customers more likely to buy?

Answer: The saved evidence points to a combination of call engagement, campaign timing, customer profile and financial situation. XGBoost feature importance ranks call duration as the strongest individual predictor (0.1332). Week/quarter variables, age groups, housing-loan status, education, marital status, occupation and balance also contribute.

Why feature importance alone is not enough: importance tells us which variables helped the model make predictions, but it does not show whether a category increases or decreases subscription and it does not prove causality. EDA is needed for direction. For example, the EDA shows stronger observed response among 60+ customers and in Q1.

Important interpretation of duration: longer calls are strongly associated with successful subscription in the historical data. However, duration is only known after the call has started. It can explain why an interaction is progressing well, but it cannot be used to decide whom to call before outreach begins.

Business interpretation: successful conversion appears to depend on both *who the customer is* and *how/when the campaign interaction occurs*. The bank should therefore avoid a single-variable targeting strategy.

Evidence source: 03_XGBoost_Modeling.ipynb - Top 20 Feature Importance; 01_Bank_Marketing_EDA.ipynb - Duration, Age Group and Quarter subscription analyses.

10.6 Which features should the bank focus on?

Answer: For pre-call targeting, focus on actionable customer-profile, financial and campaign-timing variables.

These include age/profile information, job/education context, balance, housing/personal-loan status, and timing variables such as quarter/week.

Why these features: they are supported by both the predictive model and the EDA, and most importantly they can be known before the call. This makes them operationally useful for deciding which customers to contact and when.

What not to use for initial targeting: call duration should not be a pre-call selection feature even though it has the highest XGBoost importance. Using it to decide whom to call would rely on information that does not exist at decision time.

Recommended next model: train and validate a dedicated pre-call model without duration. That model would better represent the real deployment scenario and would give the bank an interpretable customer-priority score before outreach.

Evidence source: 03_XGBoost_Modeling.ipynb - Feature Importance; 01_Bank_Marketing_EDA.ipynb - customer and campaign analyses.

10.7 Recommended Call-Center Strategy

Priority	Recommendation	Reason
1	Use XGBoost probability to rank customers before outreach.	Combines many predictive signals instead of relying on a single segment.
2	Prioritize high-probability customers in stronger historical segments.	Adds business context from EDA to model ranking.
3	Use customer profile, balance/loan status and campaign timing for pre-call decisions.	These variables are available before contact and are supported by the analysis.
4	Use duration only as an in-call engagement signal, not a pre-call targeting feature.	Duration is highly predictive but unavailable before the call starts.
5	Track precision, recall, F1 and conversion rate in addition to accuracy.	The 7.24% positive class makes accuracy alone insufficient.
6	Save mean 5-fold CV accuracy in the next training run.	Closes the only direct documentation gap against the stated success metric.

Overall business conclusion: The project supports using XGBoost as a customer-prioritization system rather than as a yes/no sales guarantee. The strongest practical strategy is to combine predicted probability with actionable customer and campaign information, while keeping the model aligned with information available at the time a call decision is made.

11. Limitations

- Random Forest and XGBoost holdout results were produced from different train-test splits, so the comparison is not a perfectly controlled same-sample benchmark.
- The saved 5-fold CV results report F1 rather than average CV accuracy.
- The positive subscriber class is strongly imbalanced.
- Feature importance is predictive, not causal.
- High segment conversion rates should be interpreted together with segment size.
- Call duration is not available before contact and should not drive initial customer selection.

12. Notebook Reference Map

Final Reference Name	Purpose in the Project
01_Bank_Marketing_EDA.ipynb	Data quality, EDA, subscription patterns and customer-segment evidence
02_Random_Forest_Modeling.ipynb	Random Forest modeling, class balancing, GridSearch and evaluation
03_XGBoost_Modeling.ipynb	XGBoost modeling, GridSearch, Optuna, evaluation and feature importance
04_Model_Comparison_and_Business_Analysis.ipynb	Final RF-vs-XGB comparison, business interpretation and consolidated conclusions

These standardized names are used throughout the report and should match the final project submission structure.